# IGCSE 0607 Practice Questions — E2.7

## Sequences

**Total Questions:** 27 — **Total Marks:** 159 Marks

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## Table of Contents & Quick Navigation


- **E2.7** Sequences
  → **Paper 2 (Non-Calculator)** *(12 Qs)* · **Paper 4 (Calculator)** *(15 Qs)*


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**1** Here are the first five terms of a number sequence.

7 11 15 19 23

**(a)** Write down the next term in this sequence.

..... **[1]**

**(b)** Find an expression, in terms of $n$, for the $n$th term of this sequence.

..... **[2]**

**2** Here are the first four terms of a sequence.

5 11 17 23

**(a)** Find the next term of this sequence.

..... **[1]**

**(b)** Find an expression for the $n$th term of this sequence.

..... **[2]**

**(c)** Explain why 150 is not a term in this sequence.

..... **[1]**

**3** Here are the first four terms of a sequence.

5 12 19 26

**(a)** Write down the next term in the sequence.

..... **[1]**

**(b)** Find an expression for the $n$th term of the sequence.

..... **[2]**

**(c)** Explain why 150 is not a term in this sequence.

..... **[1]**

**4** Here are the first four terms of a sequence.

5 11 17 23

**(a)** Find the next term in this sequence.

..... **[1]**

**(b)** Find an expression for the $n$th term of this sequence.

..... **[2]**

**(c)** Explain why 150 is not a term in this sequence.

..... **[1]**

**5** **(a)** Find the $n$th term of the sequence.

5 9 13 17 21

..... **[2]**

**(b)** Find the $n$th term of the sequence.

2 6 18 54 162

..... **[2]**

**6** The first four terms of a sequence are

7 11 15 19

**(a)** Write down the next two terms of the sequence.

..... **[1]**

**(b)** Find an expression for the $n$th term of this sequence.

$n$th term = ..... **[2]**

**(c)** Explain why 102 is not a term in this sequence.

..... **[2]**

**7** **(a)** Sequence $A$ is a linear sequence.

5 9 13 17 ...

**(i)** Find the next two terms of sequence $A$.

..... **[1]**

**(ii)** Find the $n$th term of sequence $A$.

..... **[2]**

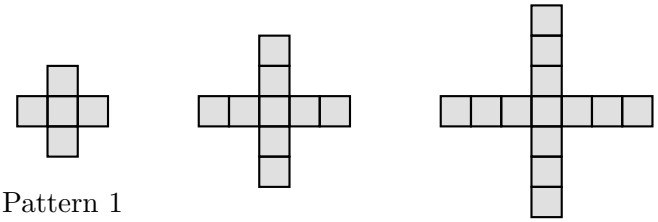
**(b)** Sequence $B$ is a geometric sequence.

2 6 18 54 ...

Find the $n$th term of sequence $B$.

..... **[2]**

**8** The diagram shows a sequence of patterns made from small squares.



Pattern 1



Pattern 2



Pattern 3

**(a)** Find the number of squares in Pattern 4.

..... **[1]**

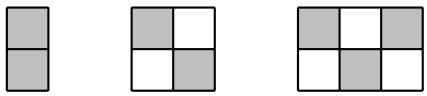
**(b)** Find an expression, in terms of $n$, for the number of squares in Pattern $n$.

..... **[2]**

**(c)** Which pattern has exactly 141 squares?

Pattern ..... **[2]**

**9** The diagram shows the first three patterns in a sequence made from congruent grey and white squares.



Pattern 1 Pattern 2 Pattern 3

**(a)** Complete the table for Pattern 4 and Pattern 5.

| Pattern number ($n$) | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Number of grey squares | 2 | 2 | 3 | | |
| Number of white squares | 0 | 2 | 3 | | |
| Total number of squares | 2 | 4 | 6 | | |

**[2]**

**(b)** Write down an expression, in terms of $n$, for the total number of squares in Pattern $n$.

..... **[1]**

**(c)** Pattern $k$ has a total of 94 squares.
Find the value of $k$.

$k =$ ..... **[2]**

**10** The first four terms of a sequence are

$$-1 \qquad 4 \qquad 13 \qquad 26$$

**(a)** Find an expression for the $n$th term of the sequence.

..... **[3]**

**(b)** Find the value of $n$ when the $n$th term is 433.

$n =$ ..... **[3]**

**11** The first five terms of a quadratic sequence $T$ are shown below.

5 12 23 38 57

**(a)** Write down the next term in sequence $T$.

..... **[1]**

**(b)** Find an expression, in terms of $n$, for $T_n$, the $n$th term of sequence $T$.

$T_n =$ ..... **[3]**

**(c)** Sequence $U$ has $n$th term $U_n = 3 \times 2^{n-1}$.
Find the value of $T_6 - U_6$.

..... **[2]**

**12** The first five terms of a sequence, $T_n$, are given below.

4 14 34 68 120

**(a)** By finding the first, second and third differences, show that the $n$th term of the sequence is a cubic expression in $n$.

**[2]**

**(b)** Find an expression for $T_n$, the $n$th term of the sequence.

$T_n =$ ..... **[5]**

**13** Here are the first five terms of a sequence.

3 9 17 27 39

Find an expression, in terms of $n$, for the $n$th term of this sequence.

..... **[3]**

**14** Here are the first five terms of a sequence.

4 11 22 37 56

**(a)** Work out the next term of the sequence.

..... **[1]**

**(b)** Find an expression for the $n$th term of the sequence.

..... **[3]**

**15** The $n$th term of a sequence is $u_n = an^2 + bn$.

**(a)** Given that $u_1 = 5$ and $u_2 = 14$, write down two simultaneous equations in $a$ and $b$.

.....

..... **[2]**

**(b)** Solve these simultaneous equations to find the value of $a$ and the value of $b$.

$a =$ ..... , $b =$ ..... **[3]**

**16** **(a)** The $n$th term of a sequence is $n^2 - 3n + 5$.
Work out the first three terms of this sequence.

1st term = ..... , 2nd term = ..... , 3rd term = .....
**[2]**

**(b)** The first four terms of another sequence are

2 9 20 35

Find an expression for the $n$th term of this sequence.

..... **[3]**

**17** The first four terms of a sequence, $T$, are

2 9 22 41

**(a)** Find an expression for the $n$th term, $T_n$.

$T_n =$ ..... **[3]**

**(b)** A second sequence is given by the formula $U_n = 2^n + 3$.
Work out the value of $U_5$.

$U_5 =$ ..... **[1]**

**(c)** Find the value of $n$ for which $U_n = 131$.

$n =$ ..... **[2]**

**18** The first four terms of a sequence are

$$-1 \qquad 2 \qquad 9 \qquad 20$$

**(a)** Find the $n$th term of this sequence.

..... **[3]**

**(b)** The $n$th term of another sequence is given by $3^n - 2$.
Find the value of the term in this sequence that is closest to 1000.

..... **[3]**

**19** The first five terms of a sequence are

2 9 20 35 54 ...

**(a)** Find the next term.

..... **[1]**

**(b)** Find the $n$th term of the sequence.

..... **[3]**

**(c)** The $k$th term of the sequence is 464.
Find the value of $k$.

$k =$ ..... **[3]**

**20** Consider the following three sequences, $P$, $Q$ and $R$.
Sequence $P$:

$$2 \quad 9 \quad 28 \quad 65 \quad 126 \quad \ldots$$

Sequence $Q$:

$$3 \quad 8 \quad 15 \quad 24 \quad 35 \quad \ldots$$

Sequence $R$:

$$\frac{2}{3} \quad \frac{9}{8} \quad \frac{28}{15} \quad \frac{65}{24} \quad \frac{126}{35} \quad \ldots$$

**(a)** Find the $n$th term of sequence $P$.

..... **[2]**

**(b)** Find the $n$th term of sequence $Q$.

..... **[2]**

**(c)** Write down the $n$th term of sequence $R$.

..... **[1]**

**(d)** Find the 12th term of sequence $R$, giving your answer as a fraction in its simplest form.

..... **[2]**

**21** Here are the first five terms of a sequence.

2 9 28 65 126

**(a)** Find the next term of this sequence.

..... **[1]**

**(b)** Find an expression for the $n$th term of this sequence.

..... **[3]**

**(c)** The $k$th term of the sequence is 8001.
Find the value of $k$.

$k =$ ..... **[3]**

**22** The table shows the first five terms of sequences $A$, $B$ and $C$.

| $n$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| Sequence $A$ | 2 | 5 | 10 | 17 | 26 |
| Sequence $B$ | 1 | 8 | 27 | 64 | 125 |
| Sequence $C$ | $-1$ | 3 | 17 | 47 | 99 |

**(a)** Find the $n$th term of sequence $A$.

..... **[2]**

**(b)** Find the $n$th term of sequence $B$.

..... **[1]**

**(c)** Use your answers to parts (a) and (b) to find an expression, in terms of $n$, for the $n$th term of sequence $C$.
Give your answer in its simplest form.

..... **[2]**

**(d)** Determine which term in sequence $A$ has a value of 10001.

$n =$ ..... **[2]**

**23** **(a)** The first five terms of a sequence are

$$-1 \quad 8 \quad 33 \quad 80 \quad 155$$

Use the method of differences to find an expression for the $n$th term of this sequence.

..... **[4]**

**(b)** An exponential sequence has $n$th term $P_n = 5 \times 3^{n-1}$.
The $k$th term of this sequence is 3645.
Work out the value of $k$.

$k =$ ..... **[2]**

**(c)** The $n$th term of sequence $T$ is $T_n = \dfrac{2n-1}{3n+2}$.
Write down the value that $T_n$ approaches as $n$ becomes very large.

..... **[1]**

**24** The first five terms of a cubic sequence $A$ are shown below.

2 10 30 68 130

**(a)** Use the method of differences to find an expression, in terms of $n$, for $A_n$, the $n$th term of sequence $A$.

$A_n =$ ..... **[4]**

**(b)** Sequence $B$ is defined by $B_n = A_{n+1} - A_n$.
Show that $B_n = 3n^2 + 3n + 2$.

**[2]**

**(c)** Find the value of $n$ for which $B_n = 272$.

$n =$ ..... **[2]**

**25** **(a)** A sequence has the following first four terms:

1 15 53 127

Use the method of differences to find an expression for the $n$th term.

$n$th term = ..... **[4]**

**(b)** Another sequence, $V$, has its $n$th term defined by

$$V_n = \frac{n^2 + 1}{2^n}$$

**(i)** Find the values of $V_1$, $V_2$ and $V_3$.
Give each fraction in its simplest form.

$V_1$ = ..... , $V_2$ = ..... , $V_3$ = ..... **[2]**

**(ii)** Work out the exact value of $V_5 - V_4$.

..... **[3]**

**26** **(a)** The first five terms of a sequence are

4 5 14 37 80

Use the method of differences to find an expression for the $n$th term.

..... **[4]**

**(b)** The $n$th term of a different sequence is given by $T_n = a \times 2^n + b$.
The first term of this sequence is 10 and the third term is 34.

Find the values of $a$ and $b$.

$a =$ ..... , $b =$ ..... **[4]**

**(c)** Calculate the 12th term of the sequence described in part (b).

..... **[2]**

**27** The first five terms of three sequences, A, B and C, are shown below.

| | | | | | |
|---|---|---|---|---|---|
| Sequence A: | 5 | 9 | 13 | 17 | 21 |
| Sequence B: | 2 | 6 | 20 | 50 | 102 |
| Sequence C: | $\frac{5}{2}$ | $\frac{3}{2}$ | $\frac{13}{20}$ | $\frac{17}{50}$ | $\frac{7}{34}$ |

**(a)** Find the $n$th term of sequence A.

..... **[2]**

**(b)** **(i)** Work out the 6th term of sequence B.

..... **[1]**

**(ii)** Find an expression for the $n$th term of sequence B.

..... **[4]**

**(c)** **(i)** Work out the 6th term of sequence C, giving your answer as a fraction in its simplest form.

..... **[1]**

**(ii)** Find an expression for the $n$th term of sequence C.

..... **[2]**